

Real-Time Streaming Data Pipeline for a Flight Booking System

- **Objective**

Design a real-time streaming data pipeline for a flight booking system using GCP BigQuery, Spark Streaming, and Kafka/GCP Pub/Sub.

- **Requirements**

- **Data Sources:**

- Two Kafka/GCP Pub-Sub Topics:
- Topic 1: Booking Data
- Topic 2: Payment Data

- **Data Processing:**

- Use Spark Streaming to read data from both topics.
- Persist booking events in state until the corresponding payment event arrives.

- **Data Joining and Insertion:**

- Join booking and payment events when they match (i.e., payment event arrives after the booking event for the same booking).
- Insert the joined record into a BigQuery table.

- **State Management:**

- Clean up the state once the record is inserted into BigQuery.

- **Event Handling:**

- Discard payment events that arrive before their corresponding booking events.

- **BigQuery BI Engine Insights:**

- Use BigQuery BI Engine and prepare dashboards for below statements
 - Understand how revenue changes over time (Yearly, Monthly)
 - Find out which payment methods are most popular
 - Identify the most frequented flight routes and destinations

❖ Sample Events and BigQuery Schema:

Booking Event

```
{  
  "booking_id": "B123",  
  "customer_id": "C456",  
  "flight_id": "F789",  
  "booking_time": "2023-04-01T10:30:00Z",  
  "origin": "LAX",  
  "destination": "JFK",  
  "amount": 300  
}
```

Payment Event

```
{  
  "payment_id": "P123",  
  "booking_id": "B123",  
  "payment_time": "2023-04-01T10:45:00Z",  
  "payment_method": "Credit Card",  
  "payment_status": "Completed"  
}
```

BigQuery Target Table Schema

booking_id (STRING)
customer_id (STRING)
flight_id (STRING)
booking_time (TIMESTAMP)
payment_id (STRING)
payment_time (TIMESTAMP)
payment_method (STRING)
payment_status (STRING)
origin (STRING)
destination (STRING)
amount (NUMERIC)

→ Deliverables

- ◆ Correct implementation of real-time streaming and state management.
- ◆ Accuracy in joining booking and payment events.

- ◆ Effective use of BigQuery for data storage and analysis.
- ◆ Innovation in deriving business insights from the data.
- ◆ Clarity and completeness of the documentation.